



Society of Petroleum Engineers

SPE-229707-MS

Study on the Influence Mechanism of Fiber Properties on the Strength of Fiber-Particle Composite Temporary Plugging Layer

Feng Zhao, Weihua Chen, and Ji Zeng, Engineering Technology Research Institute of Southwest Oil & Gas Field Company, PetroChina, Chengdu, Sichuan, China; Yukie Tanino, School of Engineering, University of Aberdeen, Aberdeen, UK; Xinyuan Tang, Tingting He, Jinsui Li, Rui He, and Tao Wang, Engineering Technology Research Institute of Southwest Oil & Gas Field Company, PetroChina, Chengdu, Sichuan, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229707-MS](https://doi.org/10.2118/229707-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Temporary plugging and diversion fracturing technology is an effective method for developing shale oil and gas reservoirs. However, temporary plugging often fails due to insufficient shear strength of the temporary plugging layer. It has been proven that mixing fibers and particles to form the temporary plugging layer can effectively improve its mechanical strength, but current research results are mostly qualitative analyses. In this paper, we present a DEM-based model to predict the shear strength of a particle-fiber mixture temporary plugging layer. The rolling resistance model simulates interactions among particles and between particles and fibers. Fibers consist of bonded particles with a uniform diameter of 0.5 mm. The influence of fiber addition on the mechanical strength is analyzed from a mesoscopic perspective through force chains, and the effects of fiber concentration (0.1%-10%), length (3 mm-25 mm), and friction coefficient (0.2-1.6) are quantitatively studied. Adding fiber improves the mechanical strength of the temporary plugging layer. For example, with 1% fiber concentration, strength increases by 35.8% compared to pure particles. While fiber addition reduces force chain strength from 820 N (no fibers) to 630 N (1% fibers), the number of force chains increases significantly from 170 to 570, compensating for the strength loss. Strength initially increases and then decreases with higher fiber length and concentration. Maximum shear strength occurs at 0.7% fiber concentration and 20 mm length. Higher fiber concentrations and lengths reduce shear strength due to decreased particle-to-particle contact. The fiber friction coefficient increases strength initially and stabilizes at 1, having the greatest impact on mechanical strength, followed by fiber concentration, while fiber length has the least effect. This work constructs a quantitative correlation between fiber characteristics and the shear strength of composite temporary plugging layers, offering technical guidance for optimal fiber material selection and the innovation of plugging agents used in diversion fracturing processes.

Keywords: temporary plugging, DEM, fiber particles mixing, force chain

Introduction

Temporary plugging diversion technology involves pumping a temporary plugging agent into the formation, where it gradually compacts from a loose to a dense state under fluid pressure (Zhao and Tanino et al., 2024; Ren et al., 2023). This compaction creates a plugging layer that alters the stress distribution in the formation, controls the growth of existing fractures, and promotes the formation of new fractures (Zhao et al., 2024). After the process is complete, the temporary plugging agents dissolve into small molecular substances due to formation temperature and fluid interactions (Chen et al., 2024). As these small molecules flow back to the surface, the original fracture conductivity is restored (Guo et al., 2024). The plugging agents are primarily composed of polyester and polylactic acid, which break down into environmentally benign byproducts, ensuring minimal environmental impact (Zhou et al., 2025). This technology has been successfully applied in geothermal energy extraction (Xie et al., 2024), unconventional reservoir development (Chang et al., 2024), and carbon capture and storage (CCS) (Han et al., 2024).

Field engineers assess the effectiveness of temporary plugging by observing the wellhead pressure response when a temporary plugging agent is pumped into the formation. However, there are often cases in the field where the plugging capacity is insufficient, leading to temporary plugging failure (Wang et al., 2020). One of the main reasons is insufficient mechanical strength, where the plugging layer cannot withstand the pressure differential and subsequently fails, leading to a sudden drop in plugging pressure (Zhao and Tanino et al., 2024).

The most common method for creating a temporary plugging layer involves injecting particles and fibers into the formation. The mechanical strength of this layer depends on several geometric and mechanical properties of the fibers, such as concentration, length, diameter (Xu et al., 2024; Yang et al., 2019; Zou et al., 2025).

Previous studies have demonstrated that fiber parameters have a significant impact on the mechanical strength of temporary plugging materials based on experiment. Wang et al. (2020) and Li et al. (2020) reported that increasing fiber content leads to enhanced strength, while Ma et al. (2019) observed that strength first increases and then decreases with higher fiber content, indicating the existence of an optimal range. Regarding fiber diameter, Bao et al. (2019) found that larger diameters contribute to higher mechanical strength. In terms of fiber length, Wang et al. (2020) showed that longer fibers exhibit a more pronounced reinforcement effect.

While previous studies have experimentally examined the mechanical strength of fiber–particle composites, they lack robust predictive models. Key parameters, such as fiber friction coefficient, remain underexplored due to material constraints and the absence of quantitative analysis. In this work, a shear strength model based on the discrete element method (DEM) is developed to evaluate the effects of fiber concentration, length, and friction coefficient. The relative influence of each parameter is quantified, and an optimization framework for fiber design is established and validated in carbonate reservoirs of the Sichuan Basin, showing favorable application outcomes.

Materials and method

Basic property of particle and fiber

The particles and fibers used in this study are provided by LMPS Technology Company. Physical images of particles and fibers are shown in the Fig. 1. The composition of the particles is mainly polyester, and the composition of the fiber is mainly PVA.

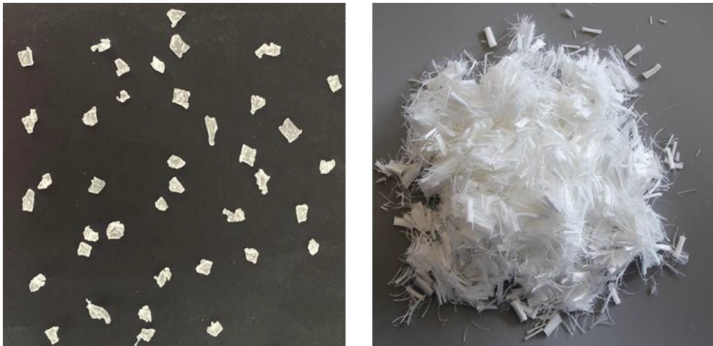


Figure 1—particle and fiber used in this study

The characteristic parameters of particles mainly include geometric parameters and mechanical parameters. The parameters provided by the manufacturer are shown in the Table 1. Since the parameters of the particles vary from batch to batch, they need to be re-tested in the study. The geometric parameters are the equivalent particle size, length-diameter ratio and sorting ability which are obtained by Image J software. The mechanical parameters, including the particle friction coefficient and Young's modulus, were determined using a friction coefficient instrument and nano-indentation instrument, respectively, as described in our previous study (Zhao and Tanino et al., 2024.)

Table 1—the property of particle

Particle size category (mm)	d_e (mm)	α	circularity	sorting	E (GPa)	μ
2 to 3	2.45	1.83	0.61	0.013	5.74	0.601

The parameters of fiber also include geometric parameters and mechanical parameters. The geometric and mechanical parameters are provide by the manufacturer as shown in Table 2.

Table 2—the property of fiber

length (mm)	diameter (um)	Young's modulus (GPa)	Friction coefficient
6	20	6.8	0.6

Model building

Shear box configuration. The numerical simulation domain was designed to replicate a laboratory shear box setup with dimensions of 200 mm × 120 mm, enclosed by eight rigid boundary walls that were sequentially labeled to facilitate operation tracking. [Fig. 2(a)].

Fiber modeling. Fibers were constructed as continuous structures discretized into a series of bonded spherical particles using a contact bond algorithm (Bao et al., 2025). Each spherical element had a uniform diameter of 0.5 mm, and the fibers were randomly positioned within the computational domain. [Fig. 2(b)].

Particle assembly. Aggregate particles were generated in the third stage using a particle exclusion approach, ensuring that their size distribution corresponded to laboratory test results and shear box characterization. [Fig. 2(c)]

Shearing process. The final simulation step involved applying shear to the packed medium under simulated formation stress using a servo-controlled mechanism. After initial vertical loading by walls 1 and 5, the shearing motion was initiated by walls 1, 2, and 8 at a controlled displacement rate of 2.4 mm/min, covering a shear distance ranging from 0 to 14 mm. The servo system dynamically adjusted wall velocities to maintain the target stress level. [Fig. 2(d)].

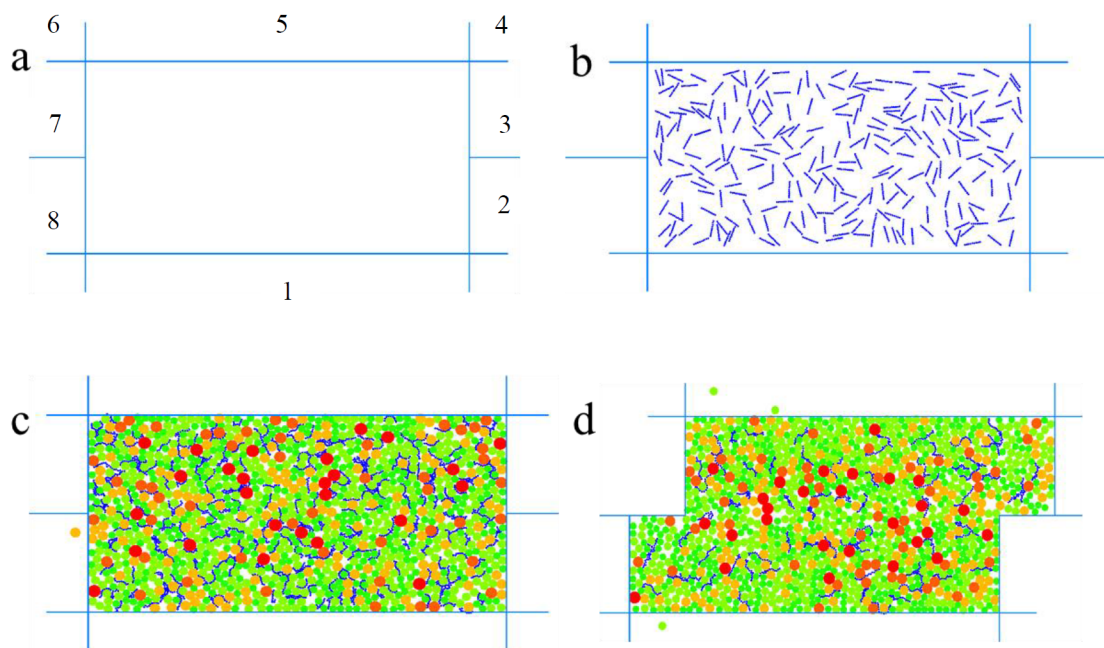


Figure 2—(a) The numerical shear box in its empty state (b) generation of fibers (c) generation of particles and fibers (d) shear

In this study, the Discrete Element Method (DEM) is employed (Zhao and Tanino et al., 2024), which is based on the core assumptions that particles are rigid disks interacting through a soft-contact model allowing slight overlap, walls apply acceleration via a servo mechanism rather than direct forces, and both velocities and accelerations remain constant within each time step.

Selection of contact models

In the simulated composite system, four primary contact types are considered: internal fiber contacts, interactions among fibers, contacts between fibers and particles, and inter-particle contacts, as illustrated in Fig. 3. To represent the rolling behavior of irregular particles while maintaining computational efficiency, a rolling resistance linear contact model is employed, following the methodology of Wensrich et al. (2012). For modeling the mechanical response of fibers, which are composed of bonded spherical elements, the linear contact bond model is utilized to capture the strength and flexibility intrinsic to fiber structures (Rabiei et al., 2023). This model effectively accounts for fiber deformation under tensile, shear, and bending loads. The standard linear contact model is applied to represent both the fiber–particle and fiber–fiber interactions. A detailed description of each contact mechanism is provided in the subsequent subsections.

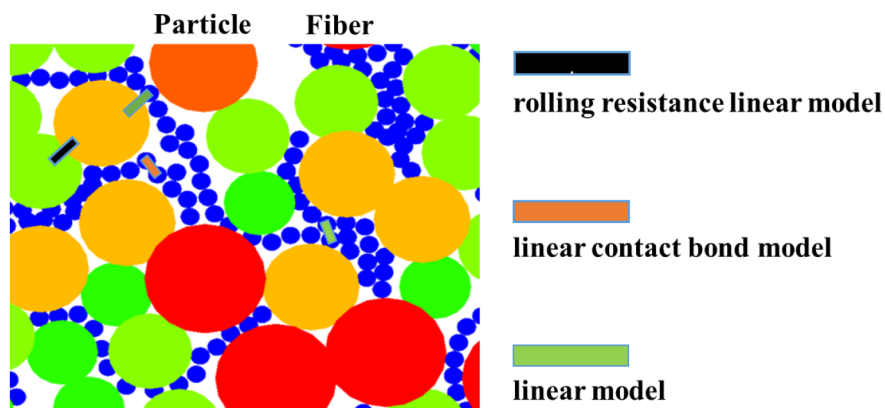


Figure 3—Assignment of the contact model in this study

(1) Linear model

The contact force is decomposed into a normal component, F_n , and a tangential component, F_t . F_n is given by:

$$F_n = \begin{cases} k_n g_n & g_n \leq 0 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Here, k_n defines the stiffness normal to the contact surface, and g_n quantifies the degree of overlap between contacting particles. The formulation for the tangential force F_t is as follows:

$$F_t = \begin{cases} |F_t^*| & \text{if } |F_t^*| \leq \mu F_n \\ \mu F_n & \text{otherwise} \end{cases} \quad (2)$$

$$F_t^* = (F_t)_0 - k_t g_s \quad (3)$$

Here, $(F_t)_0$ refers to the starting value of the tangential contact force, k_t characterizes the tangential stiffness, and g_s indicates the variation in relative displacement between contacting surfaces.

(2) Linear contact bond model

The overall contact force is further divided into two components: a normal force ($\bar{F}_n$) and a tangential force ($\bar{F}_s$). The normal force can be described as:

$$\bar{F}_n = (\bar{F}_n)_0 + \bar{k}_n \bar{g}_n \quad (5)$$

In this expression, $(\bar{F}_n)_0$ represents the initial value of the normal force, $\bar{k}_n$ indicates the normal stiffness coefficient, and $\bar{g}_n$ refers to the relative displacement between contacting particles.

$$\bar{F}_s = (\bar{F}_s)_0 - \bar{k}_s \bar{g}_s \quad (6)$$

Here, $(\bar{F}_s)_0$ refers to the starting value of the tangential contact force, while $\bar{k}_s$ defines the stiffness in the tangential direction, and $\bar{g}_s$ captures the relative shear displacement between particles.

(3) Rolling resistance linear model

The contact model is based on linear model, with the addition of a rolling resistance mechanism. The rolling resistance moment, M_r , is calculated as:

$$M_r = \begin{cases} |M_r^*| & \text{if } |M_r^*| \leq \mu_r \bar{R} f_n \\ \mu_r \bar{R} f_n & \text{otherwise} \end{cases} \quad (7)$$

$$M_r^* = (M_r)_0 - k_r \Delta \theta_r \quad (8)$$

where μ_r denotes the coefficient of rolling resistance, R_e represents the effective contact radius, $(M_r)_0$ is the initial torque resisting rotation, k_r is the rotational stiffness coefficient, and $\Delta \theta_r$ signifies the angular increment between the contacting particles. The expression for normal stiffness k_n is:

$$k_n = \frac{\pi r^2 E^*}{r_A + r_B} \quad (9)$$

where r_A and r_B are the radii of the two particles and $r = \min(r_A, r_B)$. The rolling resistance stiffness, k_r , is given by:

$$k_r = k_t \bar{R}_a^2 \quad (10)$$

Model verification

To verify the reliability of the numerical model, a comparison was conducted with results from direct shear experiments carried out using a laboratory device developed at Southwest Petroleum University (Fig.3a).

In the experiment, Samples were loaded into a shear box with dimensions of 100 mm × 100 mm. A vertical load equivalent to 100 kPa was applied to the plugging layer through weighted compression. After the initial loading phase, the sample was subjected to shear deformation at a constant displacement rate of 2.4 mm per minute.

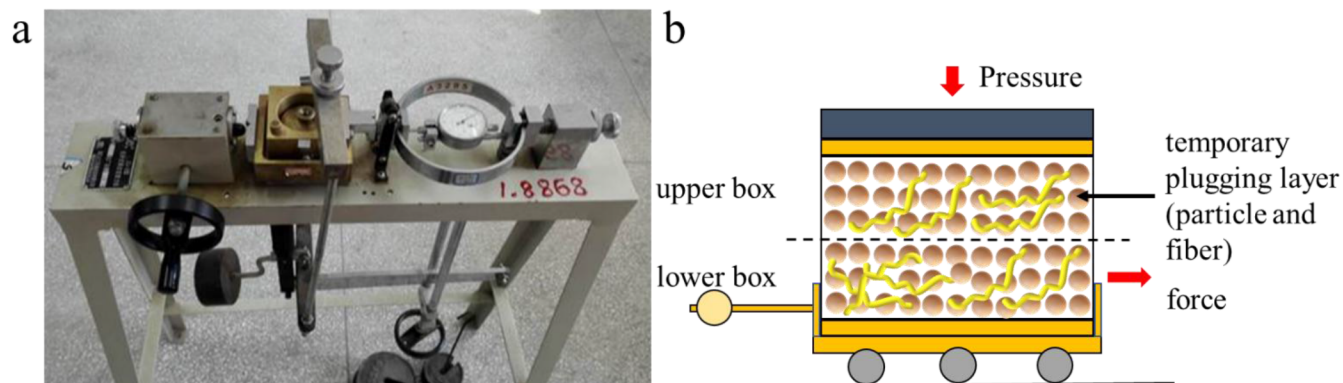


Figure 3a—Direct shear apparatus (a) photograph (b) schematic diagram

Results and discussion

Analysis of stress–strain characteristics of direct shear test

As shown in Fig. 4, the numerical model developed in this study demonstrates a high degree of consistency with the experimental results. In particular, the simulated stress–displacement response of the fiber–particle system (black line in Fig. 4) closely matches the experimental curve (orange line in Fig. 4), confirming the accuracy of the DEM model in capturing the shear mechanical behavior of the composite plugging material.

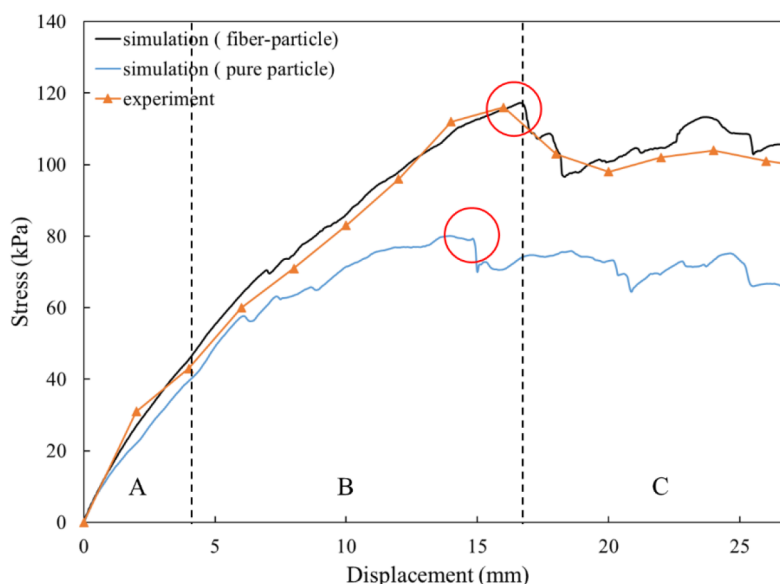


Figure 4—shear stress versus displacement of temporary plugging layer

The stress–displacement curve of the fiber–particle system can be divided into three distinct stages. Stage A (0–5 mm) corresponds to a linear increase in stress, representing the initial formation of particle contacts. In stage B (5–16.6 mm), stress increases nonlinearly as the internal structure becomes denser and the fibers begin to stretch and constrain the system, thereby enhancing its load-bearing capacity. The peak stress of 114.2 kPa is reached at a displacement of 16.59 mm (highlighted by the red marker in Fig. 5), indicating the

formation of a stable force-bearing network between particles and fibers. In stage C, partial structural failure occurs, accompanied by a slight decrease and fluctuation in stress, reflecting the onset of shear softening and structural reorganization.

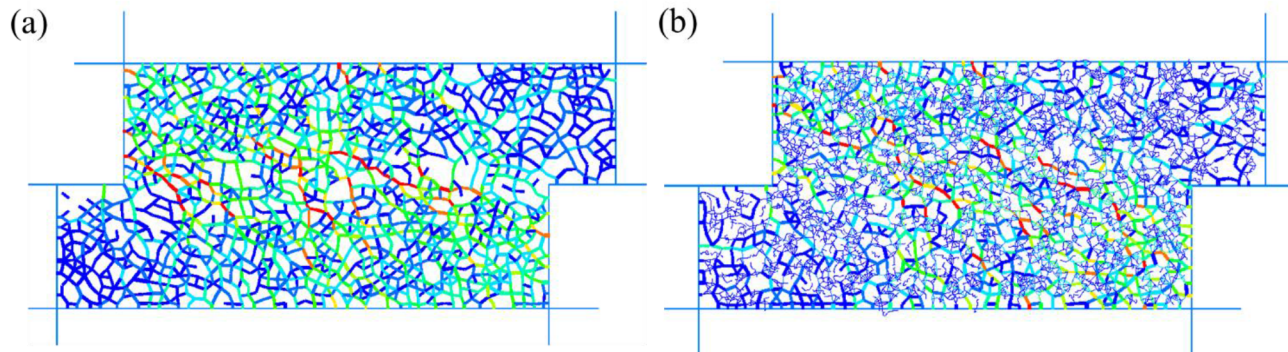


Figure 5—Force chain structures in temporary plugging layer (a) particle (b) fiber-particle

In contrast, the pure particle system (blue line in Fig. 4) exhibits a slower rate of stress growth and reaches its peak stress of 80.0 kPa at a displacement of 14.11 mm (also marked in red in Fig. 4). This peak is significantly lower—by approximately 35%—than that of the fiber–particle system, and failure occurs at an earlier stage, indicating lower structural stability and a greater tendency toward collapse. These results clearly demonstrate that the inclusion of fibers significantly enhances the ultimate load-bearing capacity of the plugging structure, delays the onset of failure, and improves the ductility and deformation resistance during shear loading.

Statistical analysis of the force chain characteristics within the temporary plugging structure under peak stress conditions reveals significant differences between the fiber-particle composite system and the pure particle system in terms of force chain architecture (Fig. 5). The identification criteria for force chains were based on our previously published literature (Zhao and Tanino et al., 2024). The results demonstrate that although the average strength of individual force chains in the pure particle system (842 N) is notably higher than that in the fiber-particle system (625 N), the latter exhibits superior performance in terms of force chain quantity and structural complexity. Specifically, the fiber-particle system forms 572 effective force chains, far exceeding the 172 observed in the pure particle system. Among these, long force chains with lengths greater than 3 account for 341 in the fiber-particle system, compared to only 106 in the pure particle system. This indicates that while the incorporation of fibers does not enhance localized load-bearing capacity, it significantly improves the density of the load-bearing network and its topological extensibility within the system.

From a structural mechanics perspective, the abundance of long force chains facilitates the construction of more stable load-bearing pathways, enabling broader stress distribution and dissipation throughout the system. This mechanism reduces the risk of localized stress concentration and enhances shear resistance. In contrast, although the pure particle system exhibits higher localized strength, its sparse and loosely connected force chain network renders the overall structure more susceptible to rapid failure under high stress. In conclusion, the load-bearing capacity of a system is governed not only by the strength of individual force chains but more critically by their quantity and spatial distribution. The fiber-particle system achieves superior macroscopic strength performance and delayed failure precisely by establishing a denser and more extended load-bearing network.

Effect of fiber parameters on the strength of temporary plugging layer

The influence of different fiber parameters on the mechanical strength of the temporary plugging layer is illustrated in Fig. 6. As shown in Fig. 6 (a), the mechanical strength increases with fiber concentration at

first and then decreases. When the fiber concentration is 0.4%, the peak stress reaches 98.3 kPa, rising to 106.5 kPa at 0.6%. The maximum strength of 122.2 kPa appears at 0.7% concentration, representing a 52.3% increase compared to the case without fiber addition. However, further increasing the concentration leads to a decline in strength, dropping to 106 kPa at 2.0%. This behavior is mainly due to the disruption of the stable particle skeleton structure by excessive fiber content, which weakens the overall load-bearing capacity. A similar trend is observed for fiber length, as illustrated in Fig. 6 (b). The strength initially rises with increasing fiber length and then falls after exceeding the optimal value. The highest strength of 121 kPa occurs at a fiber length of 17 mm. Fibers that are too short fail to provide sufficient structural reinforcement, while overly long fibers may cause entanglement or uneven distribution, impairing the integrity of the plugging layer. Fig. 6 (c) shows that the friction coefficient has a more significant impact on strength. As the friction coefficient increases from 0.6 to 1.0, the peak stress rises markedly from 104.82 kPa to 141.32 kPa, an increase of 34.82%. When the coefficient is further raised to 1.4, the strength slightly decreases to 138.84 kPa, suggesting that the strengthening effect of friction gradually stabilizes beyond a certain point.

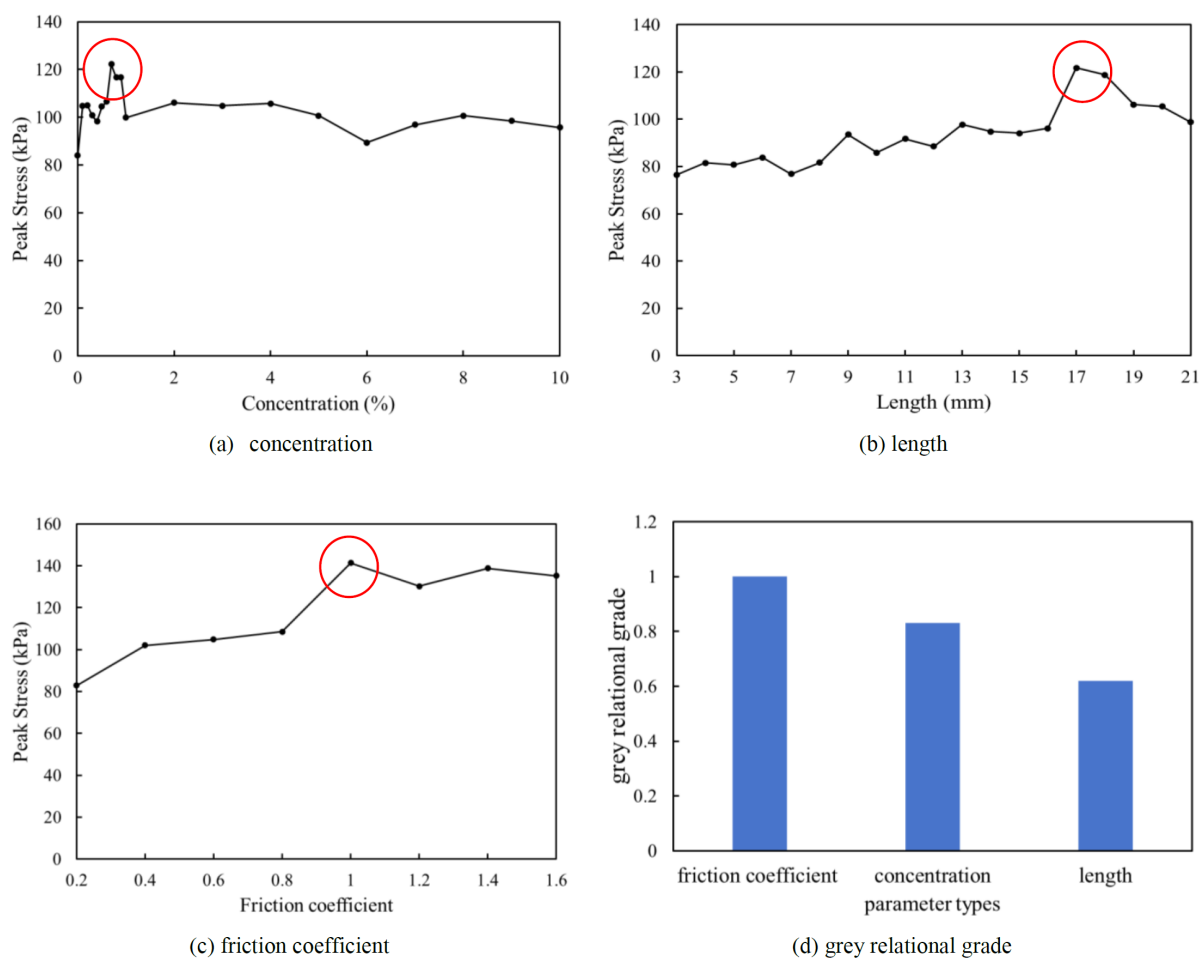


Figure 6—The effect of fiber parameters on mechanical strength (a) concentration (b) length (c) friction coefficient (d) grey relational grade

Finally, the grey relational analysis in Fig. 6 (d) evaluates the relative importance of each parameter. The results indicate that the friction coefficient has the greatest influence on the mechanical strength of the temporary plugging layer, followed by fiber concentration, with fiber length having the least effect.

Field-Ready Predictive Model

In the previous sections, discrete element method (DEM) simulations were employed to systematically investigate the effects of fiber concentration, length, and friction coefficient on the mechanical strength of the temporary plugging layer. The simulations revealed clear nonlinear influence patterns and identified key controlling factors. However, despite the accuracy of such numerical methods, their application in field settings is often limited by the need for substantial computational resources and complex model setup.

To enhance the practical applicability of the findings, a regression-based predictive model was developed based on the DEM results, aiming to provide a fast and convenient tool for field engineers. The model uses fiber–matrix friction coefficient (μ), fiber concentration (C), and fiber length (l) as input variables and incorporates second-order terms to capture the nonlinear behavior. The resulting empirical formula is given as:

$$\sigma = 94.18 + 23.35\mu + 4.53\mu^2 + 15.07C - 2.76C^2 - 2.21l + 0.15l^2 \quad (11)$$

where σ is the predicted mechanical strength of the plugging layer. The model achieves a coefficient of determination of $R^2=0.797$ indicating a good level of predictive accuracy. This equation enables rapid strength estimation based on accessible field parameters without requiring complex simulations, offering a practical reference for the design and optimization of temporary plugging treatments.

Field Application

To validate the practical applicability of the proposed predictive model, a carbonate reservoir well in the Sichuan Basin (hereinafter referred to as Well MX019) was selected as a case study. The reservoir has a porosity of 7.87% and a permeability of 1.02 mD, with a total stimulated interval length of 1358 m and a measured depth of 5782 m. A temporary plugging acid fracturing process was employed, using a self-generating acid prepared followed by a gelling acid system (Zeng et al., 2025). Temporary plugging optimization was conducted in stages 1, 4, and 7, where the minimum horizontal principal stress differences were 15 MPa, 10 MPa, and 14 MPa, respectively. To enhance fracture diversion and reservoir stimulation efficiency, a composite plugging system composed of fibers and particles was designed, targeting a minimum plugging strength of 15 MPa.

According to pressure responses from fracturing treatment curves (blue circle in Fig. 7), the bottomhole pressure increased by 15.3 MPa after plugging, indicating that the composite plugging system successfully established effective blockage and met the design requirements. The plugging material formulation consisted of 1.0% 8 mm fibers (with a friction coefficient of 0.96), 0.5% 10/40 mesh particles, and 0.5% 100 mesh powder.

By inputting these parameters into the proposed empirical prediction model (formula 11), the predicted mechanical strength is calculated to be 126.2 kPa under the model's default vertical pressure of 100 kPa. Assuming a linear relationship between mechanical strength and vertical stress (Saberian et al., 2021), the predicted strength under the actual in-situ vertical pressure of the well is approximately 16.8 MPa, thereby confirming the accuracy and field applicability of the proposed model.

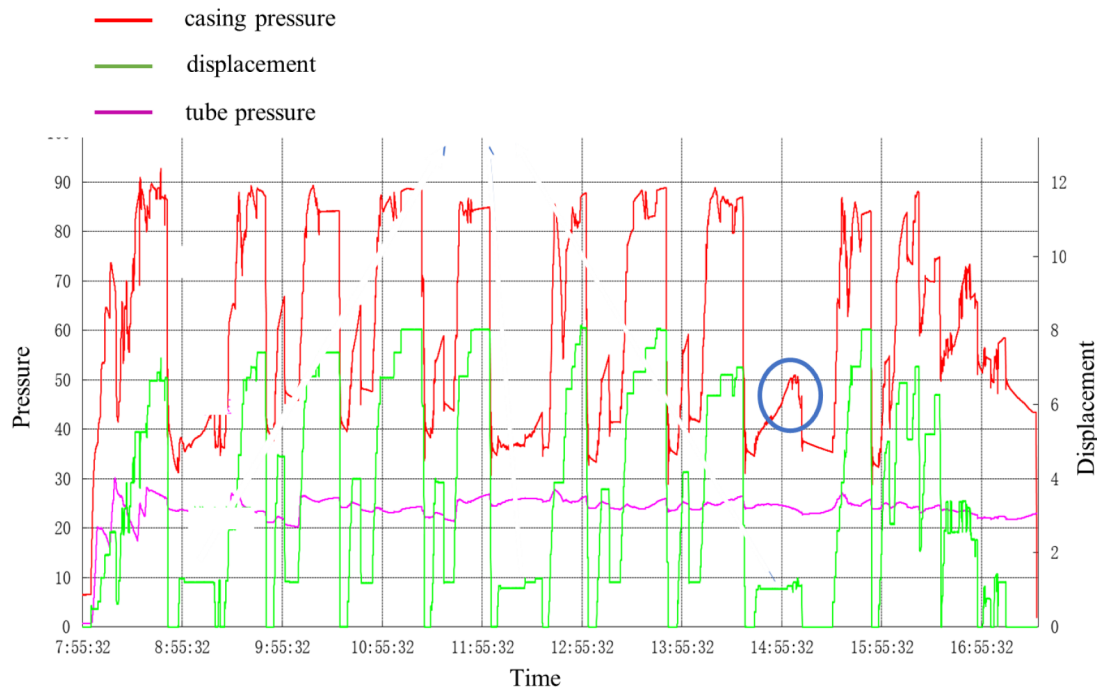


Figure 7—Operation curve of Well MX109 (Red: casing pressure; Green: displacement; Purple: tubing pressure)

Conclusion

This study establishes a mechanical model of fiber-particle composite temporary plugging layers based on the Discrete Element Method (DEM), systematically investigating the evolution of their mechanical behavior during shear failure. By analyzing peak stress, the influence mechanisms of fiber parameters—including concentration, length, and friction coefficient—on mechanical strength were clarified, and an analytical model of the layer's strength was developed. The research outcomes have been successfully applied to carbonate reservoir development in the Sichuan Basin. The main conclusions are as follows:

- The shear failure process of fiber-particle plugging layers can be divided into three stages: (i) linear stress increase, (ii) nonlinear stress increase (with a slower growth rate relative to shear displacement), and (iii) failure. The results show that the incorporation of fibers enhances the mechanical strength of the plugging layer by 35.8%.
- Fibers improve the mechanical strength by increasing both the number and length of force chains within the plugging layer. The strength exhibits a non-monotonic trend with fiber length and concentration, increasing initially and then decreasing (optimal parameters: length = 17 mm, concentration = 0.7%). It increases with the fiber-particle friction coefficient and then levels off (critical value: $\mu = 1$). The relative influence of the parameters on strength follows the order: friction coefficient > fiber concentration > fiber length.
- Based on the DEM simulation data, an analytical model for the mechanical strength of fiber-particle composite plugging layers was developed ($R^2 = 0.797$), and its reliability was validated using field data from Well MX019 in the Sichuan Basin.

Reference

- Bao, D., Qiu, Z., Zhao, X., 2019. Experimental investigation of sealing ability of lost circulation materials using the test apparatus with long fracture slot. *J. Petrol. Sci. Eng.* **183**(0): 106396.
- Bao, X., Bao, Z., Shen, J., Chen, X., Cui, H., 2025. An optimization DEM modelling method for soil-fibre undrained cyclic loading tests model based on the influence range of fibre. *Comput. Geotech.* **177**: 106846.

- Chen, X., Lu, X., Liu, P., Du, J., Liang, C., Huang, Q.S., Zhu D.W., Liu F., 2024. A critical review of key points in temporary plugging fracturing: Materials, injection, temporary plugging, and design. *Geoenergy Sci Eng.* **240**: 212981.
- Wensrich, C.M., Katterfeld, A., 2012. *Rolling friction as a technique for modelling particle shape in DEM. Powder Technol.* **217**: 409-417.
- Chang, X., Wang, X. Y., Yang, C. H., Guo, Y. T., Wei, K., Hu, G., Jiang, C. B., Li, Q., Dou, R.J., 2024. Simulation and optimization of fracture pattern in temporary plugging fracturing of horizontal shale gas wells. *Fuel.* **359**: 130378.
- Han, L., Shi, X., Ni, H., Zhang, W., Gao, Q., 2024. Review of CO₂ fracturing and carbon storage in shale reservoirs. *Energy. Fuels.* **38**(17): 15913-15934.
- Li, J. J., Yang, H. X., Qiao, Y., Zhen, F., Chen, W. B., Jiang, H. Q., 2018. Laboratory evaluations of fiber-based treatment for in-depth profile control. *J. Petrol. Sci. Eng.* **171**: 271-288.
- Ma, H. Y., Wen, Q. Z., Luo, M. L., Yu, T. T., 2018. Performance evaluation and optimization of temporary plugging agent used in diverting fracturing. In: International Field Exploration and Development Conference. Singapore: Springer Singapore, 2018: 95-107.
- Ren, J. C., Zhao, F., Guo, J. C., 2023. Influence mechanism and optimization of the plugging capacity of temporary plugging zones with different particle sizes based on pore structure. *Geoenergy Sci Eng.* **230**: 212269.
- Rabiei M, Samea P, Shadi A, Ghoreishi-Madiseh, S.A., 2023. A discrete element analysis for general failure behavior of basalt. *Int. J. Rock. Mech. Min.* **167**: 105394.
- Saberian M, Li J, Perera S T A M, Zhou, A., Roychand, R., Ren, G., 2021. Large-scale direct shear testing of waste crushed rock reinforced with waste rubber as pavement base/subbase materials. *Transportation Geotechnics*, **28**: 100546.
- Wang, B., Zhou, F.J., Yang, C., Wang, D.B., Yang, K., Liang, T.B., 2020. Experimental study on injection pressure response and fracture geometry during temporary plugging and diverting fracturing. *SPE J.* **25**(2), 573-586.
- Xie, Z., Han, D., Li, J., Li, K., 2024. A State-of-the-Art Review of Hydraulic Fracturing in Geothermal Systems. *Sustainability.* **16**(24): 11087.
- Xu, W., Jiang, F., Wang, L., Liao, Y., Qiu, S., 2024. Experimental study on plugging behavior of multitype temporary plugging agents in hydraulic fractures. *SPE J.* **29**(12): 6756-6774.
- Yang, C., Zhou, F. J., Feng, W., Tian, Z. D., Yuan, L. S., Gao, L. Y., 2019. Plugging mechanism of fibers and particulates in hydraulic fracture. *J. Petrol. Sci. Eng.* **176**: 396-402.
- Zhao, F., Tanino, Y., Guo, J. C., He, R., Ren, J. C., Zeng, J., Chen, W. H., Huang, L K., 2025. *Bed strength in sheared beds of mono-and bi-disperse particles: Dependence on geometrical and mechanical properties of constituent particles. Powder Technol.* **449**: 120286.
- Zhao, F., Guo, J. C., Ren, J.C., 2024. Research on the influence and mechanism of particle geometry and mechanics on the strength of temporary plugging zone. In: SPE Gas & Oil Technology Showcase and Conference. Onepetro. SPE-219370-MS.
- Zhou, Q., Luo, Z., Fu, H. R., Chen, K. Y., Guo, Y., Lu J. Z., 2025. Research and application progress of temporary plugging agent for acidification fracturing: A review. *Geoenergy Sci Eng.* **247**: 213600.
- Zou, Y. S., Li, D. Y., Yang, C., Li, C.Y., Zhang, S. C., Zou, L. Q., Ma, X. F., 2025. *Temporary plugging agent transport behavior within visualized multi-fracture created during TPDF in a horizontal well: An experimental study. Petrol Sci.*
- Guo, J., Zhao, F., Ren, J., 2024. Experimental study on plugging behavior of temporary plugging particles during dynamic increase of fracture width. In: International Petroleum Technology Conference. IPTC. IPTC-23174-MS.
- Zeng, M. Y., Zhou, H.Y. 2025. Investigation of Acid-Etching Evolution in Natural Fractures of Carbonate Reservoirs Under High Closure Stress. *SPE J.* **30**(01): 98-110.